Solutions and teaching notes, chapter 8

Teacher use. The following answers concern the tasks not solved in the lesson. First have pupils make their approach explicit; provide graduated help before the solution. The durations are estimates to be calibrated in class.

Diagnostic and remediation

1. 4. *Action:* Add, then divide by the number of values.
2. Yes: positive numbers with sum 1. *Action:* Put over a common denominator.
3. 9. *Action:* Keep the brackets before squaring.
4. 0,4. *Action:* Use the complement.
5. 5. *Action:* Write net gain = amount received minus amount paid.
6. 0,5. *Action:* Check the square of the answer.

Investigation

Gains (-1; 0; 1) and (-3; 0; 3) : expectation 0 in both cases, variances $\frac{2}{3}$ and 6. The second is three times more spread out in terms of standard deviation.

Direct applications

1. Values 0,1,2 ; probabilities $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$. $P(X \leq 1) = \frac{3}{4}$.
2. $E(G) = \frac{10}{5}$ = 2. For a stake m, $E(G - m) = 2 - m$; fair stake 2.
3. $E(X) = 0$, $E(X^2)$ = 1, variance 1 and standard deviation 1.
4. $E(Y) = -1$, $V(Y) = 16$, $\sigma(Y) = 4$.
5. $\frac{2\sigma}{\sqrt{n}}$ = 1; interval [4; 6].

Exercises: methods and answers

Exercise 8.1 · A distribution to construct. Values 0,1,2; each obtained from two faces, probabilities $\frac{1}{3}$. $P(X < 2) = \frac{2}{3}$. Note the remainder 0 for 3 and 6.

Exercise 8.2 · Two variance calculations. $E(X) = -\frac{1}{2} + 1 = \frac{1}{2}$. $E(X^2) = 1 + 4 = 5$; $V = 5 - \frac{1}{4} = \frac{19}{4}$, $\sigma = \frac{\sqrt{19}}{2}$. Direct: $(25\frac{}{4})(\frac{1}{4}) + (\frac{1}{4})(\frac{1}{2}) + (\frac{49}{4})(\frac{1}{4}) = \frac{76}{16} = \frac{19}{4}$.

Exercise 8.3 · A fair game. $E(G) = \frac{15}{5}$ = 3, so m=3. $E(G^2) = \frac{125}{5} = 25$; $V(G) = 25 - 9$ = 16, standard deviation 4, unchanged by the stake.

Exercise 8.4 · Same mean, different risk. $E(X) = E(Y) = 1$; $V(X) = 0$; $E(Y^2) = 10$, $V(Y) = 9$. X is certain, Y is spread out. A player's preferences cannot be ranked solely from the expectation.

Exercise 8.5 · Linearity without independence. $E(X) = \dfrac{1+2+3+4+5+6}{6} = \dfrac{7}{2}$; $E(Y) = 7 - E(X) = \dfrac{7}{2}$. Since $X + Y = 7$, its expectation is 7, the sum of the two expectations. The events $\{X = 1\}$ and $\{Y = 6\}$ are identical: their intersection has probability $\dfrac{1}{6}$, while the product of their probabilities is $\dfrac{1}{36}$. The variables are not independent. Linearity still holds without this hypothesis.

Exercise 8.6 · A minimum loss. Write $X - a = (X - \mu) + (\mu - a)$, expand and then use $E(X - \mu) = 0$. The square $(a - \mu)^2$ is minimal, zero only for $a = \mu$. The mean minimises the mean squared deviation.

Exercise 8.7 · Letter frequencies. Convert the text to lower case, keep the characters for which `isalpha()` is true, then count the "a"s. Divide this number by the number of letters if this is non-zero. For a text with no letters, return `None`, since the frequency is undefined. For `Aba !`, two "a" out of three letters give $\dfrac{2}{3}$; the other two tests give `None`. The supplied code `frequence_a` follows this convention.

Exercise 8.8 · Delayed review. Dist.: 0 , 1, 2 with probabilities $\dfrac{1}{4}, \dfrac{1}{2}, \dfrac{1}{4}$. Expectation 1 ; moment of order 2: $\dfrac{1}{2} + 1 = \dfrac{3}{2}$, variance $1\dfrac{1}{2}$. The four paths are counted explicitly.

CQFD, Exercise 28 p. 306, all questions. $E(X) = (-10) \times 0{,}35 + (-5) \times 0{,}15 + 0 \times 0{,}4 + 20 \times 0{,}1 = -2{,}25 = -\dfrac{9}{4}$. A negative expectation is compatible with a possible positive value.

CQFD, exercise 31 p. 306, all questions; use the distribution from exercise 28. $E(X^2) = 100 \times 0{,}35 + 25 \times 0{,}15 + 400 \times 0{,}1 = 78{,}75$. Thus $V(X) = 78{,}75 - (-2{,}25)^2 = 73{,}6875 = \dfrac{1179}{16}$, then $\sigma(X) = \dfrac{\sqrt{1179}}{4} \approx 8{,}6$. Do not round the expectation before calculating the variance.

MCQ: answers and reasons

MCQ 1.

1. B. Partition of the sample space.

2. A. $0 + \dfrac{4}{4}$.

3. D. König-Huygens.

4. C. Positive square root.

5. B. Linearity.

MCQ 2.

1. C. A translation adds no spread.

2. A. Absolute value of the factor.

3. D. A variable may be negative or have a mean that it never takes.

4. B. A common finite sample space is enough.

5. C. $\frac{4}{10} = 0{,}4$.

True or false

1. False: it is a mean of squares with positive weights.

2. False: the constant 1 and an equally likely variable $-2,\ 4$ have the same mean.

3. True: the deviations from the new mean stay the same.

Differentiated teaching. If a pupil is stuck on a technical point, suggest only the first step (domain, factor or reference sample space), then remove the help. For an independent student, ask for a second method and the handling of a boundary case. Orally, allow an initial rephrasing in the language the pupil is fluent in, then require a precise mathematical sentence in French.